

AA411

Fault Isolator Test Plan

Team Name: ROFASO

Project Name: Damage Assessment Protocol (D.A.P.)

Sub-system Name: Fault Isolator

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1 Objective of Test Plan

The goal of this procedure is to test that the fault isolator can accurately determine wing, rudder, or unknown damage to the Seaglider independently from other DAP systems.

i Scope

- Sub-system: Fault Isolator
- Type: Code verification
- Outputs: Percentage Difference between fault cases and mission data

2 Test Parameters and Variables

i Parameters

Parameters	Description	Value	Source
Mission Data	The data from the Seaglider mission	All	SG.github
Simulated digital twin (DT) Data	The data from the mission representing a functional DT	All	DT Code

ii Variables

Variable	Description	Units
Independent Variables (Control Inputs)		
100 % Nominal	Fault Case with no modifications to the Seaglider	Mission data units
50 % Wing	Fault Case with 50 percent of the wing span remaining	Mission data units
0 % Rudder	Fault Case with 0 percent of the rudder span remaining	Mission data units
Dependent Variables (State Variables)		
PitchAngle	Pitch of Seaglider	(deg)
RollAngle	Roll of Seaglider	(deg)
Heading	Orientation of Seaglider	(deg)
PitchAngleRate	Rate of change of Pitch of Seaglider	(deg/s)
RollAngleRate	Rate of change of Roll of Seaglider	(deg/s)
HeadingRate	Rate of Change of Orientation of Seaglider	(deg/s)

3 Test Matrix

Full Test Matrix:

Test ID	Mission Input	DT Input 1	DT Input 2	DT Input 3	Score 1	Score 2	Score 3	Package Configuration
P1R1	MI_{N5}	MI_{N5}	MI_{W11}	MI_{R8}	Nominal	Rudder	Wing	N5 / R8 / W11
P1R2	MI_{R8}	MI_{N5}	MI_{W11}	MI_{R8}	Rudder	Wing	Nominal	N5 / R8 / W11
P1R3	MI_{W11}	MI_{N5}	MI_{W11}	MI_{R8}	Wing	Rudder	Nominal	N5 / R8 / W11
P2R1	MI_{N5}	MI_{N5}	MI_{W12}	MI_{R8}	Nominal	Rudder	Wing	N5 / R8 / W12
P2R2	MI_{R8}	MI_{N5}	MI_{W12}	MI_{R8}	Rudder	Wing	Nominal	N5 / R8 / W12
P2R3	MI_{W12}	MI_{N5}	MI_{W12}	MI_{R8}	Wing	Rudder	Nominal	N5 / R8 / W12
P3R1	MI_{N5}	MI_{N5}	MI_{W11}	MI_{R9}	Nominal	Rudder	Wing	N5 / R9 / W11
P3R2	MI_{R9}	MI_{N5}	MI_{W11}	MI_{R9}	Rudder	Wing	Nominal	N5 / R9 / W11
P3R3	MI_{W11}	MI_{N5}	MI_{W11}	MI_{R9}	Wing	Rudder	Nominal	N5 / R9 / W11
P4R1	MI_{N5}	MI_{N5}	MI_{W12}	MI_{R9}	Nominal	Rudder	Wing	N5 / R9 / W12
P4R2	MI_{R9}	MI_{N5}	MI_{W12}	MI_{R9}	Rudder	Wing	Nominal	N5 / R9 / W12
P4R3	MI_{W12}	MI_{N5}	MI_{W12}	MI_{R9}	Wing	Rudder	Nominal	N5 / R9 / W12
P5R1	MI_{N6}	MI_{N6}	MI_{W11}	MI_{R8}	Nominal	Rudder	Wing	N6 / R8 / W11
P5R2	MI_{R8}	MI_{N6}	MI_{W11}	MI_{R8}	Rudder	Wing	Nominal	N6 / R8 / W11
P5R3	MI_{W11}	MI_{N6}	MI_{W11}	MI_{R8}	Wing	Rudder	Nominal	N6 / R8 / W11
P6R1	MI_{N6}	MI_{N6}	MI_{W12}	MI_{R8}	Nominal	Rudder	Wing	N6 / R8 / W12
P6R2	MI_{R8}	MI_{N6}	MI_{W12}	MI_{R8}	Rudder	Wing	Nominal	N6 / R8 / W12
P6R3	MI_{W12}	MI_{N6}	MI_{W12}	MI_{R8}	Wing	Rudder	Nominal	N6 / R8 / W12
P7R1	MI_{N6}	MI_{N6}	MI_{W11}	MI_{R9}	Nominal	Rudder	Wing	N6 / R9 / W11
P7R2	MI_{R9}	MI_{N6}	MI_{W11}	MI_{R9}	Rudder	Wing	Nominal	N6 / R9 / W11
P7R3	MI_{W11}	MI_{N6}	MI_{W11}	MI_{R9}	Wing	Rudder	Nominal	N6 / R9 / W11
P8R1	MI_{N6}	MI_{N6}	MI_{W12}	MI_{R9}	Nominal	Rudder	Wing	N6 / R9 / W12
P8R2	MI_{R9}	MI_{N6}	MI_{W12}	MI_{R9}	Rudder	Wing	Nominal	N6 / R9 / W12
P8R3	MI_{W12}	MI_{N6}	MI_{W12}	MI_{R9}	Wing	Rudder	Nominal	N6 / R9 / W12

Table 1: Full Isolator Validation Test Matrix

Test ID	Mission Input	DT Input 1	DT Input 2	DT Input 3	Score 1	Score 2	Score 3	Package Configuration
P1R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N5 / R8 / W11
P1R2	MI_{R8}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N5 / R8 / W11
P1R3	MI_{W11}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N5 / R8 / W11
P2R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N5 / R8 / W12
P2R2	MI_{R8}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N5 / R8 / W12
P2R3	MI_{W12}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N5 / R8 / W12
P3R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N5 / R9 / W11
P3R2	MI_{R9}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N5 / R9 / W11
P3R3	MI_{W11}	$MI_{N5} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N5 / R9 / W11
P4R1	MI_{N5}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N5 / R9 / W12
P4R2	MI_{R9}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N5 / R9 / W12
P4R3	MI_{W12}	$MI_{N5} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N5 / R9 / W12
P5R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N6 / R8 / W11
P5R2	MI_{R8}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N6 / R8 / W11
P5R3	MI_{W11}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N6 / R8 / W11
P6R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Nominal	Rudder	Wing	N6 / R8 / W12
P6R2	MI_{R8}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Rudder	Wing	Nominal	N6 / R8 / W12
P6R3	MI_{W12}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R8} \pm 10\%$	Wing	Rudder	Nominal	N6 / R8 / W12
P7R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N6 / R9 / W11
P7R2	MI_{R9}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N6 / R9 / W11
P7R3	MI_{W11}	$MI_{N6} \pm 10\%$	$MI_{W11} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N6 / R9 / W11
P8R1	MI_{N6}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Nominal	Rudder	Wing	N6 / R9 / W12
P8R2	MI_{R9}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Rudder	Wing	Nominal	N6 / R9 / W12
P8R3	MI_{W12}	$MI_{N6} \pm 10\%$	$MI_{W12} \pm 10\%$	$MI_{R9} \pm 10\%$	Wing	Rudder	Nominal	N6 / R9 / W12

Table 2: Full Isolator Noise Test Matrix

4 Traceability

Requirement F-I1 is verified by test cases P1R1 through P8R3. True/False? _____

Test packages:

MI_{N5} = Nominal Dive 5

MI_{N6} = Nominal Dive 6

MI_{R8} = 0% Rudder Dive 8

MI_{R9} = 0% Rudder Dive 9

MI_{W11} = 50% Wing Dive 11

MI_{W12} = 50% Wing Dive 12

Test Runs per Package:

MI_N = Nominal Mission Input

MI_{50W} = 50% Wing Fault Mission Input

MI_{0R} = 0% Rudder Fault Mission Input

Run 1: $MI_{\text{input}} = MI_N$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$

Run 2: $MI_{\text{input}} = MI_{0R}$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$

Package	Nominal Dataset	Rudder Dataset	Wing Dataset
1	MI_{N5}	MI_{R8}	MI_{W11}
2	MI_{N5}	MI_{R8}	MI_{W12}
3	MI_{N5}	MI_{R9}	MI_{W11}
4	MI_{N5}	MI_{R9}	MI_{W12}
5	MI_{N6}	MI_{R8}	MI_{W11}
6	MI_{N6}	MI_{R8}	MI_{W12}
7	MI_{N6}	MI_{R9}	MI_{W11}
8	MI_{N6}	MI_{R9}	MI_{W12}

Table 3: Isolator Validation Test Packages

Test Run	Mission Input	Simulated DT Input 1	Simulated DT Input 2	Simulated DT Input 3	Expected Best Match
1	MI Nominal	MI Nominal	MI 50W	MI 0R	Nominal
2	MI 50R	MI Nominal	MI 50W	MI 0R	0R
3	MI 50W	MI Nominal	MI 50W	MI 0R	50W

Table 4: Isolator Validation Test Runs per Package

Run 3: $MI_{\text{input}} = MI_{50W}$; $DT_{\text{input}} = \{MI_N, MI_{50W}, MI_{0R}\}$